

1 MoE

R = routed expert, **S** = always-active shared expert, and **Dense×d** = the first d layers use a standard dense FFN.

1.1 Three Main Load-Balancing Methods

Let N be the number of experts, K the Top- K value, and T the number of tokens. Define the actual load ratio f_i and routing-probability ratio P_i for expert i as

$$f_i = \frac{1}{KT} \sum_{t=1}^T \mathbf{1}[i \in \mathcal{S}_t], \quad P_i = \frac{1}{T} \sum_{t=1}^T p_{t,i}.$$

1.1.1 A. Traditional GShard/Switch Auxiliary Loss

$$\mathcal{L}_{\text{bal}} = \alpha N \sum_{i=1}^N f_i P_i$$

This directly changes the Router through gradient updates.

1.1.2 B. DeepSeek-Style Auxiliary-Loss-Free Balancing

$$\mathcal{S}_t = \text{TopK}_i(s_{t,i} + b_i).$$

$$w_{t,i} = c \frac{s_{t,i}}{\sum_{j \in \mathcal{S}_t} s_{t,j}}, \quad i \in \mathcal{S}_t.$$

The bias b_i **only affects expert selection, not the final mixing weights**. It is updated from load statistics. Here f_i is the actual load ratio of expert i ; the target load is uniform across experts: $\bar{\mathbf{f}} = (1/N, \dots, 1/N)$, i.e. $\bar{f}_i = 1/N$. Underloaded experts receive a positive bias correction, while overloaded experts receive a negative one:

$$b_i \leftarrow b_i + \gamma \text{sign}(\bar{f}_i - f_i)$$

1.1.3 C. Kimi K3 Quantile Balancing

Kimi K3 does not use an additional load-balancing loss. Instead, it estimates the next-step expert bias directly from the Router-score margin distribution.

For each token i , compute the current biased scores $s_{i,j} + b_j^{(t)}$ and take Top- $(K + 1)$. Let the $(K + 1)$ -th largest value be the biased cutoff:

$$\alpha_i^{(t)} = \text{Top}_{K+1,j} \left(s_{i,j} + b_j^{(t)} \right).$$

Then compute the margin of each expert j relative to this cutoff using the original Router score:

$$m_{i,j}^{(t)} = s_{i,j} - \alpha_i^{(t)}.$$

The next-step, uncentered expert bias is

$$\hat{b}_j^{(t+1)} = -\text{Quantile}_{1-K/N} \left(m_{:,j}^{(t)} \right)$$

Finally, K3 also mean-centers the bias across experts.

$$b^{(t+1)} = \hat{b}^{(t+1)} - \text{mean} \left(\hat{b}^{(t+1)} \right) \mathbf{1}$$

1.2 Model Comparison

Model	Total / active parameters	MoE configuration	Router & weights	Load balancing	Dense layers	Special design
Kimi K3	2.78T / 104.2B	93 layers; $16/896R + 2S$	$s = \text{Sigmoid}(z)$; select Top-16 by $s + b$; normalize s within Top-16; scale = 1.0	Class C	First 1 layer	Stable LatentMoE: R uses the full input for routing, then compresses the input into a latent representation for weighted aggregation by Top-16 SiTU-GLU experts, followed by RMSNorm, up-projection, and addition with the Shared Expert output.
DeepSeek-V4-Pro	1.6T / 49B	61 layers; $6/384R + 1S$	$s = \sqrt{\text{softplus}(z)}$; select Top-6 by $s + b$; normalize s within Top-6; scale = 2.5	Class B; adds a lightweight sequence-wise loss	None; first three layers use Hash-MoE	The first three layers use Hash-MoE, where the expert is determined by a predefined hash function over the input Token ID; clamped SwiGLU.
GLM-5.2	744B / 40B	78 layers; $8/256R + 1S$	Sigmoid; select Top-8 by $s + b$; normalize within Top-8; scale = 2.5	Class B	First 3 layers	No additional design
MAI-Thinking-1	962B / 34.7B	78 layers; $8/512R$	Softmax gating; select Top-8; whether the selected weights are renormalized and the final scale are unspecified	Class A	Alternating	LatentMoE: routes using the full representation while transmitting only the compressed latent.
TML Inkling	975B / 41B	66 layers; $6/256R + 2S$	Sigmoid; add bias only to Routed Experts before selecting Top-6; jointly normalize the 6R and 2S scores; apply scale = $8 \times$ a learnable per-layer scale to the normalized joint aggregate	Class B	First 2 layers	The Sigmoid scores of 6 Routed Experts and 2 Shared Experts are jointly normalized; the joint aggregate receives a fixed factor of 8 and a learnable per-layer scale (this value is particularly interesting—worth inspecting).
MiMo-V2.5-Pro	1.02T / 42B	70 layers; $8/384R$	Sigmoid; select Top-8 by $s + b$; normalize s within Top-8	Class B	First 1 layer	No additional design
MiniMax-M3	428B / 23B	60 layers; $4/128R + 1S$	Sigmoid; select Top-4 by $s + b$; normalize the original s within Top-4; scale = 2.0	Class B	First 3 layers	Clamped SwiGLU.
Qwen3.5-397B-A17B	397B / 17B	60 layers; $10/512R + 1S$	Full 512-expert Softmax; select Top-10 by p ; renormalize within Top-10; scale = 1.0; the Shared Expert has a separate Sigmoid Gate	Class A	None	The Shared Expert is scored separately.
LongCat-2.0	1.6T / 48B	38 layers; $12/(768R + 128Z)$, where Z denotes a Zero-compute Identity Expert	Full 896-expert Softmax; select Top-12 by $p + b$; no Top-12 renormalization; scale = 9	Class B; adds a lightweight auxiliary loss adapted to Zero Experts	None	The 128 Z experts are zero-compute Identity Experts (MoE++—great!).
HY-3	295B / 21B	80 layers; $8/192R + 1S$	Sigmoid + expert bias; select Top-8 by $s + b$; normalize within Top-8; scale = 2.826	Class B	First 1 layer	No additional design
DeepSeek-V3	671B / 37B	61 layers; $8/256R + 1S$	Sigmoid; select Top-8 by $s + b$; normalize s within Top-8; scale = 2.5	Class B; adds a lightweight sequence-wise loss	First 3 layers	$256R \rightarrow 8$ groups; select Top-4 groups; then select Top-8 experts within them; each token visits at most 4 nodes.